PRACTICAL ASSESSMENT

Sensor Assignment

Classification and application of sensors for small mobile robots

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Declaration: We declare that this report is our own work. All information taken from textbooks, datasheets and websites has been paraphrased in our own words and acknowledged in the reference list.

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1. Introduction

A small mobile robot has to answer three practical questions while it is moving: what is around me,how far have I moved, and where am I? It cannot answer any of them without sensors. The motors and the control program can only act on the information the sensors supply, so the choice of sensors decides what the robot is actually able to do. A robot fitted only with a contact switch can react after it has already hit something, while the same robot fitted with a distance sensor can slow down before it makes contact.

Sensors used on mobile robots are normally described using three pairs of terms. The first pair separates sensors that measure the robot itself from sensors that measure the world outside it. Siegwart and Nourbakhsh (2004) call these proprioceptive and exteroceptive sensors; we use the terms internal and external, as set out in the assignment brief. The second pair separates sensors that produce the energy they measure from sensors that only receive energy that is already present, which is the difference between active and passive. The third pair separates a reading taken relative to the robot itself from a reading taken relative to a fixed reference in the world, which is the difference between local and global.

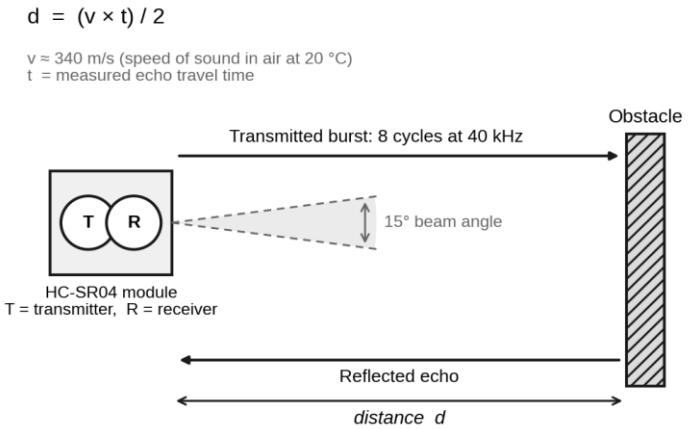
These classifications are not just labels. They predict how a sensor will behave on a real robot. An internal, local sensor builds its answer up step by step from a known starting point, so its error grows the further the robot drives, while an external, global sensor calculates a fresh answer every time and therefore does not drift (Borenstein, Everett & Feng, 1996).

We investigated five sensors that are small, light and inexpensive enough to be fitted to a small mobile robot. For each one we state what it measures, explain how it works, apply all three classification pairs with our reasons, identify the robot function it serves, and comment on how well it suits a small platform. We then summarise all five in one comparison table and use the conclusion to show how they can be combined so that the weaknesses of one sensor are covered by the strengths of another.

2. Sensor investigations

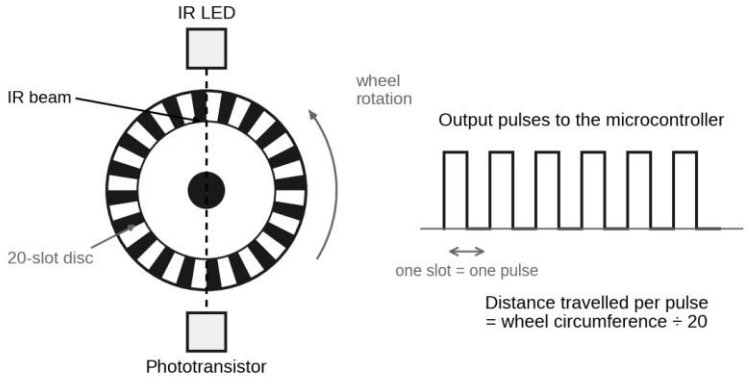
We have set out each sensor using the investigation template given in the brief. We chose these five so that they use genuinely different sensing methods (reflected sound, an interrupted light beam, inertial forces, mechanical contact and satellite radio signals) and so that between them they cover all four purpose categories

2.1 Sensor 1: Ultrasonic distance sensor

Required field	Student response
Sensor name and model	Ultrasonic ranging module, HC-SR04
Image or labelled diagram	 *Figure 1: Time-of-flight measurement using the HC-SR04. The module sends a burst of sound, waits for the echo, and the controller converts the round-trip time into distance.*
Measured quantity	The time taken for a short burst of sound to travel to a surface in front of the robot and reflect back. The controller converts this time into a distance of between about 2 cm and 4 m.
Operating principle	The controller holds the module's trigger pin high for 10 microseconds. The transmitting transducer then sends out a burst of eight cycles of sound at 40 kHz, which is above the range of human hearing. At the same moment the echo pin goes high, and it stays high until the receiving transducer picks up the returning reflection. The width of that echo pulse is therefore equal to the time the sound took to travel to the object and back. Because the sound covers the distance twice, the measured time is halved before it is multiplied by the speed of sound, which is about 340 m/s in air at 20 °C: **distance = (echo time × 340) ÷ 2** The datasheet gives the same calculation in a shorter form: distance in centimetres equals the echo time in microseconds divided by 58. The manufacturer recommends leaving at least 60 ms between readings so that the previous echo has died away before the next burst is sent (Handson Technology, n.d.).

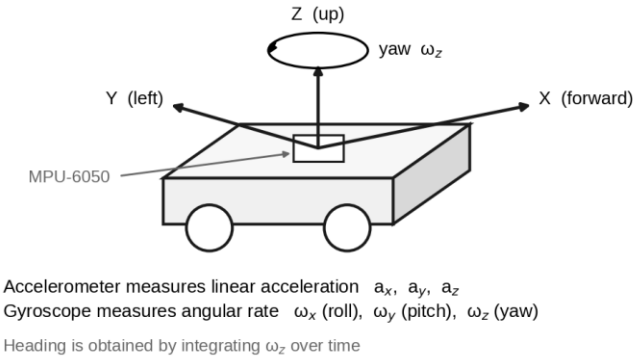
Required field	Student response
Internal or external?	**Classification: External** **Justification:** The reading describes a surface in the environment, not a condition inside the robot. It tells the robot how far away an object is, and gives no information about the robot's own speed, wheel position or battery state. Applying the guiding question in the brief, it senses an object around the robot, so it is external.
Active or passive?	**Classification: Active** **Justification:** The module generates the 40 kHz burst itself and then measures the reflection of that same burst. It is not listening to sound that was already present in the room. This is the definition of an active sensor given in the brief: the sensing method deliberately sends energy out into the environment. The fact that the module also needs a 5 V supply is not what makes it active.
Local or global?	**Classification: Local** **Justification:** The distance is measured from the face of the sensor along the direction it is pointing. The result is only meaningful in the robot's own frame of reference and says nothing about where the robot is in the room or in the world. Turning the robot changes the reading even though the robot has not moved.
Purpose category	**Purpose: Obstacle avoidance** **Explanation:** The sensor detects a surface up to roughly 4 m away, which is far enough for the robot to slow down, stop or steer around it before any contact takes place. This matches the brief's definition of obstacle avoidance rather than collision detection, because the obstacle is reported while the robot is still clear of it. A typical control rule is to keep driving while the reading stays above a set threshold, and to turn once it drops below that threshold.
Suitability for a small robot	Size and mass: the module is about 45 mm × 20 mm × 15 mm and weighs under 10 g, so it mounts easily on the front of a small chassis. Power: 5 V at less than 15 mA, which is negligible next to the drive motors. Range and accuracy: 2 cm to 400 cm with an accuracy of about 3 mm, and a beam that spreads over roughly 15° (Handson Technology, n.d.; Components101, n.d.). Cost: one of the cheapest ranging sensors available, which is why it is so common on student robots. Limitations: the 15° beam means the sensor knows something is in front of it but not exactly where in that cone. A smooth surface met at a sharp angle reflects the sound away instead of back, so a wall approached diagonally can be missed completely. Soft materials such as curtains or foam absorb the sound and also return a weak echo. There is a dead zone below 2 cm, and the 60 ms measurement cycle limits the module to roughly 16 readings per second. If several ultrasonic sensors are fitted they can pick up one another's bursts, so they should be fired one at a time.
Source(s)	Handson Technology (n.d.); Components101 (n.d.); Siegwart & Nourbakhsh (2004).

2.2 Sensor 2: Optical incremental wheel encoder

Required field	Student response
Sensor name and model	Slot-type infrared opto-coupler speed sensor module (LM393 comparator) fitted with a 20-slot encoder disc
Image or labelled diagram	 *Figure 2: An optical incremental encoder. The slotted disc turns with the wheel and chops the infrared beam, producing one pulse for every slot that passes.*
Measured quantity	The rotation of the wheel, counted as pulses. From the pulse count the controller calculates how far that wheel has turned, how far the robot has travelled and how fast it is going.
Operating principle	A thin disc with slots cut around its edge is fixed to the wheel or to the motor shaft, and it turns inside a U-shaped housing. An infrared LED sits on one arm of the U and a phototransistor sits directly opposite it on the other arm. As the disc turns, the solid parts of the disc block the beam and the slots let it through, so the phototransistor switches repeatedly between conducting and not conducting. An LM393 comparator on the module cleans this up into a square digital pulse that can be fed straight to an interrupt pin on the microcontroller (Teach Me Microcontrollers, n.d.). One slot passing the beam produces one pulse, so a 20-slot disc gives 20 pulses per wheel revolution. If the wheel is 65 mm in diameter its circumference is π × 65 = 204 mm, so each pulse means the robot has moved about 204 ÷ 20 = 10.2 mm. By counting the pulses from the left and right wheels separately the controller works out both how far the robot has driven and how much it has turned. A quadrature encoder uses two sensors placed slightly apart so that their two pulse trains are out of step. The order in which the two signals change reveals the direction of rotation as well, which a single-channel module cannot do.
Internal or external?	**Classification: Internal** **Justification:** It measures the movement of the robot's own wheel. The reading comes entirely from a part of the robot and would be exactly the same whether the robot were in an empty room or a crowded one. Siegwart and Nourbakhsh (2004) list wheel encoders as the standard example of a proprioceptive, or internal, sensor.

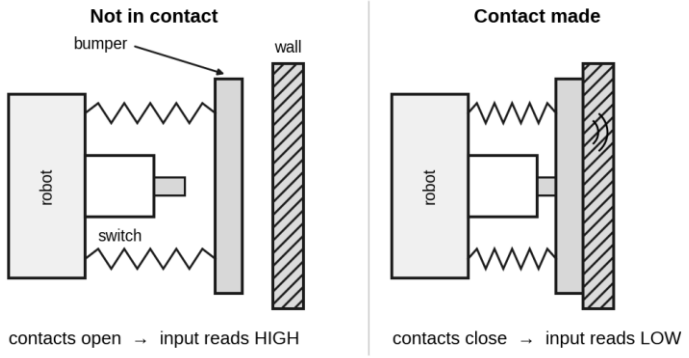
Required field	Student response
Active or passive?	**Classification: Active** **Justification:** The infrared LED on the module produces the light that the phototransistor measures. The sensing method depends on energy the sensor itself emits, and the module would give no useful signal at all if that LED were removed. Note that this classification comes from the emitted beam and not from the module needing a supply voltage.
Local or global?	**Classification: Local** **Justification:** The encoder only reports change. Position is built up by adding each small movement to the previous estimate, starting from wherever the robot happened to be switched on, so the answer is always relative to the robot's own starting point. It also means every small error stays in the estimate, and the total error grows the further the robot drives.
Purpose category	**Purpose: Dead reckoning** **Explanation:** This is the sensor that makes dead reckoning possible. The controller takes the last known position and heading, adds the distance each wheel has turned since then, and calculates the new position and heading. This matches the brief's definition exactly: estimating the current pose from a previously known pose and the measured movement. Odometry based on wheel encoders is described by Borenstein, Everett and Feng (1996) as the most widely used positioning method on mobile robots.
Suitability for a small robot	Size and mass: the module is smaller than a postage stamp and the disc is a thin plastic wheel, so the pair adds almost nothing to the robot. Power: 3.3 V to 5 V with a digital output that connects directly to a microcontroller pin, so no extra interface circuitry is needed. Accuracy: with a 20-slot disc and a 65 mm wheel the smallest movement the robot can detect is about 10 mm. Discs with more slots, or a quadrature encoder, give finer resolution. Cost: very low, and two are needed (one per driven wheel). Limitations: the encoder measures how far the wheel has turned, not how far the robot has actually moved over the ground. If the wheel slips on a polished floor, spins on loose carpet, or the robot is picked up, the encoder still reports movement that never happened. These errors add up and cannot be detected by the encoder itself, so the position estimate has to be corrected from time to time by a sensor that looks outside the robot. The disc must also be mounted squarely, because a wobbling disc produces false or missed pulses.
Source(s)	Teach Me Microcontrollers (n.d.); Borenstein, Everett & Feng (1996); Siegwart & Nourbakhsh (2004).

2.3 Sensor 3: Inertial measurement unit (accelerometer and gyroscope)

Required field	Student response
Sensor name and model	Six-axis inertial measurement unit, InvenSense MPU-6050 (commonly supplied on a GY-521 breakout board)
Image or labelled diagram	 *Figure 3: The six quantities measured by an IMU. For a ground robot the yaw rate about the vertical Z axis is the most useful, because it gives the change in heading.*
Measured quantity	Linear acceleration along three axes (selectable as ±2 g, ±4 g, ±8 g or ±16 g) and rate of rotation about the same three axes (selectable as ±250, ±500, ±1000 or ±2000 degrees per second). For a robot driving on a flat floor the most useful of these is the rate of turn about the vertical axis, known as the yaw rate.
Operating principle	The MPU-6050 is a MEMS device, which means the sensing parts are microscopic mechanical structures etched into silicon inside a chip only 4 mm × 4 mm × 0.9 mm. The accelerometer contains a tiny mass held on springs between fixed plates. When the robot accelerates, the mass lags slightly behind and moves closer to one plate, which changes the capacitance between them. The chip measures that change and converts it with a 16-bit analogue-to-digital converter. Because gravity acts on the mass in the same way, a stationary accelerometer also shows which way is down, which is how tilt is measured. The gyroscope contains a small mass that is kept vibrating. When the chip is rotated, the Coriolis effect pushes the vibrating mass sideways by an amount proportional to how fast the rotation is happening. That sideways movement is again read as a change in capacitance and converted to a number. All six readings are sent to the microcontroller over the I²C bus. The gyroscope does not give the heading directly, it gives the rate of turn, so the controller adds up (integrates) the yaw rate over time to find how far the robot has turned since it started (InvenSense, 2013).
Internal or external?	**Classification: Internal** **Justification:** The chip measures the robot's own motion and orientation. It responds to the robot accelerating and turning, and it detects nothing at all about walls, floors or objects nearby. A robot standing still next to a wall and a robot standing still in an open field give identical readings.

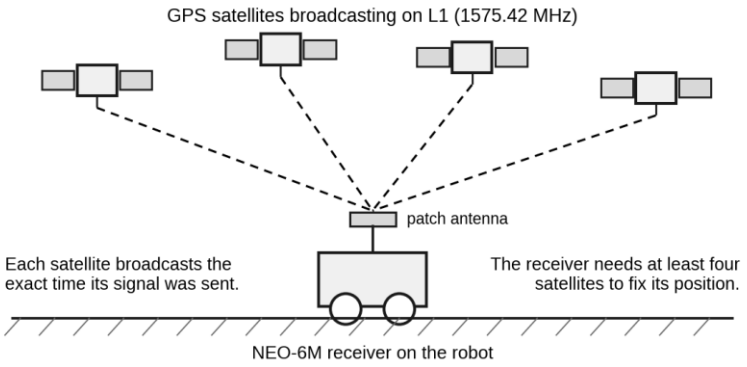
Required field	Student response
Active or passive?	**Classification: Passive** **Justification:** The IMU sends nothing out into the environment. It simply responds to forces that are already acting on it, namely the robot's own acceleration and rotation together with gravity. The brief warns specifically that needing electrical power does not make a sensor active, and this sensor is a clear example: it draws only a few milliamps, but none of that energy is emitted as a measuring signal.
Local or global?	**Classification: Local** **Justification:** The heading produced by integrating the yaw rate is measured from whatever direction the robot happened to be facing when the sensor was started. It is a change in angle rather than a bearing on a fixed world reference, so it is relative to the robot. As with the encoder, small errors accumulate and the heading estimate slowly drifts.
Purpose category	**Purpose: Dead reckoning** **Explanation:** The IMU supplies the heading part of the dead-reckoning calculation. The encoders measure how far the robot has moved, and the gyroscope measures how far it has turned. Encoders on their own estimate turning by comparing the left and right wheel counts, and that estimate is badly affected by wheel slip during a turn. The gyroscope measures the rotation of the chassis itself, so it is not affected by slip at all, and combining the two gives a noticeably better position estimate than encoders alone.
Suitability for a small robot	Size and mass: the chip is 4 mm × 4 mm × 0.9 mm and the usual breakout board is about the size of a postage stamp, so it fits anywhere on the chassis. Power: about 3.9 mA in normal operation. The chip itself runs from 2.375 V to 3.46 V, but breakout boards include a regulator so they can be supplied from 5 V. Interface: I²C, which needs only two signal wires and can share those wires with other devices. Cost: low, and it replaces two separate sensors because the accelerometer and the gyroscope are in one package. Limitations: the gyroscope has a small bias, which means it reports a slow rotation even when the robot is standing still. Because the readings are integrated, that bias turns into a heading error that grows steadily with time. The sensor should therefore be calibrated at start-up while the robot is stationary, and its heading should be corrected occasionally by an external reference. The accelerometer is also noisy on a vibrating chassis, so trying to obtain position by integrating acceleration twice is unreliable and is not normally attempted on a small robot.
Source(s)	InvenSense (2013); Siegwart & Nourbakhsh (2004).

2.4 Sensor 4: Lever-operated micro switch used as a bumper

Required field	Student response
Sensor name and model	Ultra-subminiature snap-action basic switch, Omron D2F-L (hinge-lever type), mounted behind a sprung front bumper
Image or labelled diagram	 *Figure 4: A bumper-mounted micro switch. Contact with an object pushes the bumper back against its springs, the plunger is depressed and the contacts change state.*
Measured quantity	Whether or not physical contact has taken place. The output has only two states, so the sensor answers a yes-or-no question rather than producing a measurement on a scale.
Operating principle	A light bumper bar is mounted across the front of the robot on small springs so that it can move back a few millimetres. A micro switch is fixed behind the bumper with its lever resting against it. When the robot runs into something, the bumper is pushed back, the lever presses the switch plunger, and a snap-action mechanism inside the switch flips the contacts over quickly and cleanly. When the robot reverses, the springs return the bumper and the contacts snap back. Electrically the switch is wired between a microcontroller input pin and ground, with a pull-up resistor holding the pin at the supply voltage. The input therefore reads HIGH while nothing is touching the bumper and LOW at the instant of contact. No calculation is required because the signal is already a clean digital state, although the software must ignore any change for about 10 to 20 ms after the first edge, because mechanical contacts bounce as they close. Two or three switches are normally fitted across the front so that the robot can also tell roughly whether it hit something on the left, in the centre or on the right.
Internal or external?	**Classification: External** **Justification:** What the robot learns from the switch is that an object exists in the environment at the point where the bumper is. It is true that the switch physically detects the position of the robot's own bumper, and we state that assumption here, but the bumper only moves because something outside the robot pushed it. What the robot gains is information about its surroundings, not about its own state, so we classify it as external.

Required field	Student response
Active or passive?	**Classification: Passive** **Justification:** The switch emits nothing whatsoever. The energy that operates it is supplied by the collision itself, that is, by the robot driving into the object. It only responds to something that has already happened to it, which is the definition of a passive sensor given in the brief.
Local or global?	**Classification: Local** **Justification:** The switch reports only that something is in contact with one part of the robot at this moment. It gives no distance, no direction beyond which switch was triggered, and no information about where the robot is. The result is meaningful only in the robot's immediate frame of reference.
Purpose category	**Purpose: Collision detection** **Explanation:** The switch confirms that the robot has physically made contact with an object, which is the brief's definition of collision detection. It cannot be classified as obstacle avoidance, because by the time it responds the collision has already happened. Its role on the robot is to act as the last line of defence: when the software sees the input go LOW it stops the drive motors immediately, reverses a short distance and turns away.
Suitability for a small robot	Size and mass: the switch body is about 12.8 mm × 5.8 mm × 5.7 mm, excluding the lever, and it weighs well under a gram, so several can be fitted without affecting the robot's balance. Power: none of its own. It only needs the pull-up resistor already present on most microcontroller inputs. Durability: rated for at least 1 000 000 mechanical operations and an operating range of −40 °C to +85 °C, so it will outlast the rest of the robot (Omron, n.d.). Operating force: the hinge-lever version operates at about 0.78 N in the general purpose model and 0.25 N in the low-force model, which is light enough that a small robot will trigger it before doing any damage. Cost: the least expensive of the five sensors by a wide margin. Limitations: it gives no warning at all before contact, so the robot has to be driven slowly enough that touching an obstacle is harmless. It also only protects the part of the bumper it sits behind, so an object that slips past the end of the bumper is not detected. It is therefore a supporting sensor rather than the robot's main means of dealing with obstacles.
Source(s)	Omron (n.d.); Siegwart & Nourbakhsh (2004).

2.5 Sensor 5: Satellite navigation receiver (GPS)

Required field	Student response
Sensor name and model	GPS receiver module, u-blox NEO-6M, supplied with a ceramic patch antenna
Image or labelled diagram	 *Figure 5: How a GPS receiver fixes its position. It measures how long each satellite's signal took to arrive and solves for the one point that fits all of them.*
Measured quantity	The receiver's latitude, longitude, altitude, ground speed and the exact time, all expressed against the WGS-84 world coordinate system that GPS uses.
Operating principle	A constellation of satellites orbits the Earth, each one continuously broadcasting its own orbital position together with a very precise time stamp on the L1 frequency of 1575.42 MHz. The patch antenna on the robot picks up these broadcasts. Because the receiver knows the time the signal was sent and the time it arrived, it can work out how long the signal travelled, and multiplying that by the speed of light gives the distance to that satellite. One distance places the receiver somewhere on the surface of a sphere around that satellite. Signals from at least four satellites are needed: three to find the one point where the spheres intersect, and a fourth to correct the receiver's own inexpensive clock, which is not accurate enough on its own. The NEO-6M tracks up to 50 channels and sends the calculated position to the microcontroller over a serial (UART) connection as standard NMEA sentences, once per second by default and up to five times per second if configured to do so (u-blox, 2011).
Internal or external?	**Classification: External** **Justification:** Every piece of information the receiver uses arrives from outside the robot in the form of radio signals from satellites. The module measures nothing about the robot's own wheels, motors or orientation, and a robot that is switched on but standing still still receives a full position fix.

Required field	Student response
Active or passive?	**Classification: Passive** **Justification:** This is the one we found easiest to get wrong. The receiver never transmits anything back to the satellites; it only listens. The measuring signal is produced by the satellites, not by the sensor, so under the brief's definition the module is passive. The reasoning is the same as for a camera, which uses light it did not create. A radar unit, by contrast, would be active because it sends out its own radio pulse and measures the reflection.
Local or global?	**Classification: Global** **Justification:** The position is reported as a latitude and longitude on a fixed world reference frame, not as a distance from where the robot started. This is the only one of the five sensors whose accuracy does not get worse the further the robot travels, because each fix is calculated from scratch rather than added to the previous answer. It matches the brief's guiding question directly: the robot is located using an external reference system.
Purpose category	**Purpose: Homing** **Explanation:** A home position, such as the charging point or the corner of a field, is stored in the controller as a latitude and longitude. While the robot is driving it compares its current fix with that stored point, calculates the distance and bearing between them, and steers so that the distance decreases. This matches the brief's definition of homing: the sensor helps the robot find and move towards a known target or home position. The receiver also provides the absolute correction that stops the encoder and IMU estimate from drifting, but its own task on this robot is homing.
Suitability for a small robot	Size and mass: the NEO-6M chip is 16 mm × 12.2 mm × 2.4 mm, and a complete module with its ceramic antenna is roughly the size of a matchbox, which a small robot can carry easily. Power: 2.7 V to 3.6 V at about 40 mA to 47 mA. This is the highest current draw of the five sensors and is worth planning for on a small battery. Accuracy: about 2.5 m horizontally. That is far too coarse to steer around a chair leg, but it is perfectly adequate for returning to a general area. Speed: a cold start takes about 27 s to produce the first fix and a hot start about 1 s. Updates arrive once per second by default (u-blox, 2011). Limitations: the receiver needs a reasonably clear view of the sky. It does not work indoors, under a roof or between tall buildings, and accuracy also suffers when signals reflect off nearby walls. A robot intended to work indoors would have to use something else for homing, such as an infrared beacon fitted to the docking station. The default one-reading-per-second update rate is also slow compared with the other four sensors, so it cannot be used for anything that has to react quickly.
Source(s)	u-blox (2011); Siegwart & Nourbakhsh (2004).

3. Summary comparison table

Table 1 brings all three classification pairs and the purpose category together for the five sensors we investigated above.

No.	Sensor	Internal / External	Active / Passive	Local / Global	Purpose
1.	Ultrasonic distance sensor (HC-SR04)	External	Active	Local	Obstacle avoidance
2.	Optical incremental wheel encoder (LM393 module, 20-slot disc)	Internal	Active	Local	Dead reckoning
3.	Inertial measurement unit (MPU-6050)	Internal	Passive	Local	Dead reckoning
4.	Lever micro switch on a bumper (Omron D2F-L)	External	Passive	Local	Collision detection
5.	GPS receiver (u-blox NEO-6M)	External	Passive	Global	Homing

Table 1: Classification and purpose of the five sensors investigated.

Reading down the table, our five sensors cover both sides of every classification pair and all four purpose categories in the brief. Two are internal and three are external; two are active and three are passive; four are local and one is global. Sensors 2 and 3 share the same purpose, but they arrive at it in completely different ways, one by counting wheel rotations optically and the other by measuring inertial forces, and we explain in the conclusion why a robot benefits from having both.

4. Conclusion

4.1 Comparing the five sensors

Our five sensors fall into two clear groups. The wheel encoder and the IMU are internal sensors that watch the robot itself and answer the question “how have I moved?”. The ultrasonic sensor, the micro switch and the GPS receiver are external sensors that answer “what is near me?” and “where am I?”. Neither group is much use on its own. Internal sensors give fast, smooth readings but they have no way of noticing that the robot has drifted off course, because they never look outside. External sensors can supply that correction, but they are slower, coarser, or only work under certain conditions.

The active and passive split shows a similar trade-off. The two active sensors, the ultrasonic module and the optical encoder, create the signal they measure, so they work equally well in bright light or complete darkness. The price is that they use more power and can interfere with one another, which is why several ultrasonic sensors should be fired one at a time. The three passive sensors use energy that is already there. The micro switch takes the energy of the collision itself and costs nothing to run, the IMU responds to forces already acting on the chip, and the GPS receiver only listens. Being passive keeps their power demand low, but it also means they stop working when the thing they depend on is missing, which is exactly why the GPS receiver is useless indoors.

The local and global column contains the most important difference in the whole table. The encoder and the IMU build their answer up step by step from a known starting point, so every small error stays in the estimate and the total error grows the longer the robot drives. Wheel slip and gyroscope bias are the two main causes. The GPS fix, on the other hand, is worked out fresh from the satellites every time, so its error stays at roughly 2.5 m whether the robot has driven ten metres or ten kilometres. This is the difference Borenstein, Everett and Feng (1996) describe as relative versus absolute positioning.

4.2 How combining the sensors improves the robot

Combining the five sensors removes most of the weaknesses that each one has on its own.

For safety, the ultrasonic sensor and the micro switch work as two layers. The ultrasonic sensor sees a wall several metres away and gives the robot time to slow down or turn, which handles the great majority of obstacles. It will, however, miss a smooth wall approached at a sharp angle, a curtain that absorbs the sound, and anything lying below the beam. Those are precisely the cases the bumper switch catches. Used together, the robot avoids nearly everything and fails safely on the rest, whereas either sensor alone would leave a real gap.

For position, the three positioning sensors complement one another well. The encoders measure distance travelled quickly and accurately, but their estimate of how far the robot has

turned is badly affected by wheel slip. The gyroscope measures the turn of the chassis directly and is not affected by slip at all, so using the encoders for distance and the gyroscope for heading already gives a much better estimate than either sensor could produce alone. Both, however, drift. The GPS fix does not drift, so each time a fix arrives the controller can reset its position estimate to it. The robot then relies on the fast encoder and IMU readings between fixes and on the slow but absolute GPS fix to stop the estimate wandering. The result is an estimate that is both smooth and bounded, which is more than any of the three could achieve individually.

We should add one practical qualification. Because the GPS receiver needs a clear view of the sky, a robot designed to work indoors would have to replace it. An infrared beacon mounted on the docking station would serve the same homing purpose indoors, although it would be a local rather than a global sensor and so would not correct the drift in the same way.

4.3 Closing remarks

The five sensors we selected cover all four purpose categories set out in the brief and use five genuinely different sensing methods: reflected sound, an interrupted light beam, inertial forces, mechanical contact and satellite radio signals. No single one of them is sufficient on its own. The ultrasonic sensor prevents most collisions but not all of them, the micro switch confirms the ones it misses, the encoder and the IMU keep track of movement between corrections, and the GPS receiver supplies the fixed reference that the other four cannot provide. A small mobile robot fitted with all five would be able to move about without hitting things, keep a usable estimate of its own position, and find its way back to a stored home point. That combination, rather than any one sensor, is what makes a small robot genuinely capable.

5. References

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